Macro Beta Signal v1.5–v1.7 — Model Document

Version: macro_beta_v1_7 / macro_beta_smid_v1_7 · **Status:** production · **Effective:** 2026-07-02 **Owner:** Sky Pilot Capital research · **Change control:** §9 (parameters frozen) **Code:** `Code_Repo/data/pipeline/macro_beta_clean.py`, `pipeline/macro_beta_signal.py` **Research provenance:** `Main/Planning/research/macro_beta_v2/` (00–04)

---

1. Summary

The Macro Beta Signal is a **daily, two-state US equity drawdown-defense signal**:

> **NORMAL** — no defense trigger active; ordinary market environment. **DEFENSE** — conditions historically associated with deep, macro-led equity bear markets: a bearish cycle vote (price trend / labor market / inflation momentum), and/or fast market stress (credit-spread widening, extreme realized volatility) confirmed by a broken price trend.

It is built from five well-established input families with 50+ years of history, uses fixed, fully interpretable rules (no estimated coefficients), applies every input strictly point-in-time, and switches state roughly **1.6 times per year** (v1.6: entries publish immediately; exits carry a 10-day confirmation — §3.3).

What it is for: identifying the 2000–02 / 2008–09 / 2022 class of drawdowns (–25% to –55%, unfolding over months) early enough to reduce equity beta through most of the decline. Intended uses: a standalone market-state monitor, and — in future — a beta dial for a 130/30-style portfolio implemented as an index-futures overlay.

What it is not (§7): it is not an alpha source, not fast-shock protection, and not a forecast. The 2026 research review found **no statistically robust forward-return skill across cycles for this entire model class** — v1.5's value proposition is drawdown insurance whose premium (upside given up during false alarms) is measured and published rather than hidden.

2. Background: v1 and the 2026 research review

The predecessor (`macro_beta_v1`, migrated from the legacy US_EQ_BETA_MDL) was a three-state model (high/market/low beta) voting S&P 50d/200d trend, ISM Manufacturing PMI momentum, and CPI momentum, with RSI-oversold and credit-momentum latches. A full research program (2026-07; documents 00–04 in `Main/Planning/research/macro_beta_v2/`) established:

- v1's development-window forward spread (high–low, 21d) was **t = 1.81 — not significant** over 30 years; its full-period significance rested on the single 2022 call. Its low_beta days had a 41% hit rate and a **+2.4%/yr forward excess** — most defensive days occurred in rising markets (2019 worst: 235 defensive days in a +31% year, driven by the PMI vote).
- The **PMI vote carried zero-to-negative forward information** (bearish days averaged +8.5%/yr forward). The **RSI oversold latch actively blocked defense through the 2020 crash** (latched Feb 27, held to Apr 8 while the market fell a further ~24%).
- Literature-supported upgrades (labor-market votes, credit levels, volatility gates, composites, correction channels) improved precision, timeliness, and whipsaw — but **nothing cleared a pre-registered significance bar (t ≥ 2.5), the 1973–89 quasi- out-of-sample window showed ~zero spread for every candidate, and all measured skill concentrated in the 2000s decade.** This reproduces the published record (Zakamulin; Henkel-Martin-Nardari; Goyal-Welch): timing skill of this class is a recession-episode phenomenon.

v1.5 is therefore a **surgical repair, not a new alpha claim**: each change fixes a specific documented failure. The v1 signal history is preserved in `macro_signal.beta_signal_v1_archive`.

Changes v1 → v1.5:

#	Change	Reason (documented failure)
1	Two states (defense/normal); high_beta dropped	The bullish state carried no measurable conviction
2	RSI oversold veto removed	Blocked defense through the 2020 crash
3	Realized-vol gate added (fast trigger)	2018Q4/2020/2025 fast-crash class; the one vol mechanism surviving the critique literature (conditional, not continuous)
4	Correction channel added: (credit latch OR vol gate) AND price < 10m SMA → defense	Non-recession corrections invisible to macro votes; gives day-6 COVID-class defense
5	Credit latch can force defense when cycle is not bullish	v1's mapping could never let its fastest risk-off input cause defense
6	PMI vote replaced by a labor block (claims / Sahm gap / U3 vs 12m avg)	PMI = dead weight; manufacturing-only slowdowns caused the 2011/2015/2019 false-positive class; labor data are vintage-clean with the strongest honest evidence
7	CPI momentum z vote retained unchanged	v1's best input (dev t = 1.8–2.8); made the 2022 call
8	Market series = FF total-market splice pre-1990 + ^SP500TR	Warm trend/vol statistics from 1973; longer honest history
9	CPI gap-patching switched to backward-only fill	v1 used two-sided interpolation (a look-ahead in principle)

3. Construction

3.1 Inputs

Input	Source series	Freq	History	Availability rule	Revisions
S&P 500 total return	^SP500TR (1990+); Ken French MKT_RF+RF splice (1963–89)	daily	1963+	same-day close	none
Initial jobless claims	FRED `ICSA`	weekly (Sat)	1967+	week-end + 5 days (Thu release)	minor; annual seasonal factors
Unemployment rate	FRED `UNRATE`	monthly	1948+	first Friday of following month	small (seasonal)
CPI	BLS via `raw.cpi_us`	monthly	1947+	actual historical release dates (~+13d)	headline NSA unrevised; SA seasonal-only
BBB corporate OAS	FRED `BAMLC0A4CBBB`	daily	Dec 1996+	observation + 1 business day	none
Risk-free rate	Ken French RF; DGS3MO fallback	daily	1963+	same-day	none

All inputs enter the signal only from their availability date (`merge_asof` on the availability column). The historical backfill uses current-vintage claims/UNRATE histories (revisions are small; documented caveat) — going forward, the daily pipeline captures data as published.

3.2 Derived components

- **trend_50_200_pct** = 50d MA / 200d MA – 1 (slow trend)
- **trend_10m_pct** = close / 210d MA – 1 (Faber-style 10-month trend)
- **rv21_pct10y** = 21d realized volatility (annualized), ranked within a rolling 10-year window (min 3y)
- **bbb_4_12_diff** = 20d avg – 60d avg of BBB OAS, percentage points
- **claims_ratio_12m_low** = 4-week avg claims ÷ trailing 52-week low of same
- **sahm_gap** = 3m avg U3 – trailing 12m low of same (Sahm-rule construction)
- **u3_vs_12mma** = U3 – its 12m moving average
- **cpi_mom_z3m60m** = z-score of the 3m avg of CPI MoM vs its rolling 60m distribution

3.3 Signal logic (frozen)

Cycle vote — majority of three (≥2 valid required; tie → bearish):

- **Trend**: bullish iff trend_50_200_pct > 0
- **Labor block**: bearish iff ≥2 of {claims_ratio ≥ **1.10**, sahm_gap ≥ **0.30**, u3_vs_12mma > **0**} (≥2 members must be valid)
- **Inflation (v1.7)**: bearish iff CPI YoY > **3%** AND cpi_mom_z3m60m > 0 — the momentum vote is level-gated per the Neville et al. result (rising inflation from low levels is reflation, not an equity bear signal). Verified in the 2026-07 false-alarm study to leave every episode entry unchanged, including 2022.

Fast-risk latches (hysteresis; NaN holds state):

- **Credit latch**: ON when bbb_4_12_diff > **+0.10pp** (10bp), OFF when < **0**
- **Vol gate**: ON when rv21_pct10y > **0.90**, OFF when < **0.75**

State mapping — DEFENSE if ANY of:

- `cycle_bearish` — the cycle vote is bearish. **Crater rule (v1.7, S&P only):** this path fires only when (price < 10m SMA) OR (index within **5%** of its running all-time high). Near highs, macro leads price (2000/2022 day-0 entries preserved); below the SMA the trend is already broken; in between — climbing out of a crater — a bearish macro vote alone may NOT re-fire. This removes the dominant false-alarm class (post-trough lagging labor: 2003/2009/2020-21, where unemployment keeps deteriorating for 6–12 months after the market trough). **Explicitly NOT applied to SMID**: small caps sit far below their highs chronically, and the mirror grid showed the rule *raises* SMID false-alarm cost.
- `credit_force` — credit latch ON while the cycle vote is not outright bullish
- `correction_channel` — (credit latch ON **or** vol gate ON) **and** trend_10m_pct < 0

otherwise **NORMAL**. v1.7 effect (S&P, 53 years): false-alarm cost –44% (89.5 → 49.9pp of market excess sat through), dial-0.5 premium –1.03 → –0.35%/yr, Sharpe 0.634 → 0.667, all 13 episode entries and coverages unchanged to the day. SMID (CPI gate only): false-alarm cost –38% (139.6 → 86.0pp), premium –2.13 → –1.24%/yr, Sharpe 0.427 → 0.469, all 15 episode entries unchanged. Full grids: `research/macro_beta_v2/results/false_alarm_grid*.csv`. Each defensive day records its active triggers in `defense_reasons` (e.g. `cycle_bearish(trend,labor); correction(credit,vol)`).

Exit confirmation (v1.6): the published state ENTERS defense immediately, but returns to normal only after the raw state has read normal for **10 consecutive trading days**. Rationale: diagnosis of v1.5's switching showed 39% of regime blocks lasted ≤10 days — almost all correction-channel boundary flicker (latches/trend oscillating at their thresholds). The exit confirmation merges these into single blocks: switches fell 2.54 → 1.61/yr, flicker blocks 39% → 2%, while every episode's entry date is unchanged by construction and dial-portfolio Sharpe/maxDD improved slightly. Any ENTRY delay was rejected — it broke COVID day-6 timeliness. The un-debounced state is stored alongside as `raw_state`. Pre-registered study: research log "Smoothing study" + `results/smoothing_grid.csv`. Known composition effect: the ~5pp of added defense days are rebound-tail days, which dilutes the defense-state forward-return spread without changing any entry information.

Signal history is published from **1973-01-03** (all components warm). The credit latch is structurally inactive before 1997 (no OAS series) — pre-1997 defense relies on cycle votes and the vol-gate arm of the correction channel.

3.4 SMID variant (macro_beta_smid_v1_6)

A second universe (Russell 2000 / SMID) is published alongside the S&P signal (2026-07-02; pre-registered SMID program in the research log). **Every rule and threshold above is identical.** The only differences:

- **Credit latch input: ICE BofA HY OAS** (`BAMLH0A0HYM2`) instead of BBB — the research found HY momentum carries the SMID credit signal (+13.4%/yr defense-state spread, t=1.60) while BBB carries ~none; same 10bp/0 enter/exit thresholds.
- **Evaluation asset / timeline index: Russell 2000 TR** (^RUTTR 1995-06+, hedgeable via RTY futures), back-extended pre-1995 with a parameter-free FF synthetic small-cap series (MKT_RF+SMB+RF; overlap validation: monthly corr 0.984, beta 1.02).
- **Episode reporting threshold:** ≥**20%** (–15% is routine for small caps).
- **Trend and volatility inputs remain S&P-derived** — a deliberate, tested decision: SMID's own price trend is ANTI-predictive as a bear flag (bear days +15.5%/yr forward; violent index mean reversion), and SMID-calibrated variants failed the pre-registered premium bound while losing dial Sharpe. Full grid: `research/macro_beta_v2/results/smid_grid*.csv`.

Record (full 1973+): 8 of 15 ≥20% episodes covered ≥50%, median days-to-defense 43, switches 1.80/yr; at a 0.5 dial since 1990: Sharpe 0.43 vs buy-and-hold 0.40, maxDD –37% vs –59%, premium ≈ –2.1%/yr. Documented gap (§6): relative small-cap bears without macro stress (1983-84, 2024-25) are outside the sensing range.

4. How to interpret the signal

- **NORMAL** does not mean bullish. It means no defense trigger is active. The model deliberately expresses no positive view (the review found none is warranted).
- **DEFENSE driven by** `cycle_bearish` is the recession-grade warning — the highest- conviction configuration, historically associated with the long bears.
- **DEFENSE driven only by the correction channel** is fast-risk containment: market stress with a broken trend. Expect more of these to be false alarms; they exist to buy protection against 2020-type acceleration, at a known cost in V-shaped recoveries.
- **Cadence:** ~1.7 state changes/yr (v1.7); median regime length ~2 months. A flip on a single day of data is by-design impossible for the latched components.
- **Month-end view (display layer, 2026-07):** the site also shows the daily state sampled at each completed month-end and held — a committee-cadence lens (~1.3 switches/yr historically; COVID defense would have shown Mar-2 instead of Feb-27). It is derived from, and subordinate to, the daily state, which remains the executable signal. Not a model change.
- **Reading the component board:** every threshold is fixed and published; the state is reproducible by hand. If a reading looks inconsistent with the state, check the latches (they hold between thresholds) and the availability lags (a release today enters today, not backdated).
- **The dial:** for beta-modulation use, defense maps to a reduced net beta (0.4–0.7 studied); implementation should be an index-futures overlay on an untouched portfolio, not position resizing (see the overlay study in `research/macro_beta_v2/results/overlay_study.md`).

5. Validation evidence (development window unless noted)

Full tables: research docs 03–04 and `Code_Repo/data/research/macro_beta_v2/results/`.

- **Episode record** (the product): the v1.5 defense construction covered the 1990, 1998, 2000–02, 2008 (from ~1 month in), and 2022 bears, and — unlike v1 — defended COVID 2020 from day 6 (63% of crash days). The 2018Q4 and 2025 fast corrections remain largely uncovered (§7).
- **Cost of insurance** (since 1990, published live on the site): time in defense ~25–30%; market excess return accrued during defense measures the false-alarm premium; at a 0.7 dial the historical give-up was ~0.1–0.4%/yr, scaling roughly linearly with dial depth.
- **No forward-return claim:** best dev-window High–Low spread among honest candidates t = 1.83 (bar: 2.5); 1973–89 quasi-OOS ≈ 0; skill concentrated in the 2000s decade; ~60 configurations were logged and any multiple-testing haircut nullifies the remainder. The 2020+ holdout was used exactly once.
- **Parameter robustness:** all thresholds survive ±33–50% perturbation without the headline metrics collapsing (no isolated parameter peaks).

6. Known limitations

- **Fast exogenous shocks** (2018Q4, 2025-type): largely outside the sensing range; only the vol-gate arm reacts, partially. Accepted, documented.
- **Recession-episode monetization:** measurable value concentrates in long macro-led bears; in trendless or V-shaped decades the signal is a net (small, measured) cost.
- **Credit history starts 1997**; pre-1997 behavior relies on fewer triggers.
- **Pre-1990 market series** is a total-market splice, not the S&P 500.
- **Backfilled labor/CPI histories are current-vintage** (small revisions accepted); live operation is point-in-time by construction.
- The labor block's Sahm-gap arm produced a genuine false positive in 2024 (labor-supply distortion) — the majority-vote structure absorbed it; single-member labor readings should never be traded directly.

7. What this model is NOT

- **Not an alpha model.** No forward-return skill is claimed at any horizon; none survived rigorous testing — for v1, v1.5, or any literature-supported variant.
- **Not fast-crash protection.** See §6.1.
- **Not a recession probability model**, not a rates/inflation forecast, and not a market-timing signal for entries/exits at daily granularity.
- **Not tunable in production.** See §9.

8. Operations

- **Pipeline** (droplet, daily cron Step 5): FRED/ISM/BLS raw pulls → `clean.beta_sp500_daily`, `clean.beta_credit_daily`, `clean.claims_us`, `clean.unemployment_us`, `clean.cpi_us` → `clean.beta_model_inputs_daily` → `macro_signal.beta_signal_daily` (+ `beta_episodes`, `beta_signal_stats`, `beta_dial_sim` for the website).
- **Freshness alerts** (logged by the clean flow): claims > 21d, unemployment > 70d (data-month convention), CPI > 60d, credit > 7d.
- **v1 archive:** `macro_signal.beta_signal_v1_archive` (full v1 daily history, immutable).
- **Website:** skypilotcapital.com/macro-beta reads the four `macro_signal` tables; this document is linked from the page as a PDF.

9. Change control

All thresholds and rules in §3.3 are **frozen as researched (2026-07-01)**. Any change — a threshold, an input, a mapping rule — requires a new research memo, evaluation on the pre-registered dev window through the standing harness (`Code_Repo/data/research/macro_beta_v2/harness.py`), a fresh sealed holdout, and a version bump with an updated model document. Live re-tuning against recent performance is prohibited; that failure mode is what produced v1.

Version history

Version	Date	Note
macro_beta_v1	2026-05	Legacy 3-state model migrated (PMI/CPI/trend + RSI/credit latches)
macro_beta_v1_5	2026-07-01	Two-state repair per research review
macro_beta_v1_6	2026-07-02	10-day exit confirmation on the published state (pre-registered smoothing study; user-approved). Signal content, inputs, and all thresholds unchanged
macro_beta_smid_v1_6	2026-07-02	SMID (Russell 2000) universe added: identical rules; credit latch on HY OAS; Russell 2000 TR evaluation index; ≥20% episode threshold (§3.4)
macro_beta_v1_7 / smid_v1_7	2026-07-02	False-alarm reduction (pre-registered study; user-approved): inflation vote level-gated at CPI YoY > 3% (both universes); crater rule on the cycle path (S&P only — rejected for SMID in the mirror grid). Episode entries unchanged; FA cost –44% (S&P) / –38% (SMID)

10. Key references

Faber (2007); Moskowitz-Ooi-Pedersen (2012); Zakamulin (2014); Levine-Pedersen (2016); Moreira-Muir (2017); Cederburg et al. (2020); Bongaerts-Kang-van Dijk (2020, conditional vol targeting); Harvey et al. (2018, vol targeting); Garg-Goulding-Harvey-Mazzoleni (2024, breaking bad trends); Sahm (2019); Neville et al. (2021, inflation regimes); Gilchrist-Zakrajšek (2012, EBP); Rapach-Strauss-Zhou (2010, combinations); Welch-Goyal (2008); Sullivan-Timmermann-White (1999); Harvey-Liu-Zhu (2016). Full annotated reviews: research docs 02a–02c.